



## USER GUIDE

firmware 1.2 for the M-VAVE FM-1

GHOULBOX is built on Felucca by Hügelson Instruments. M-VAVE and FM-1 are trademarks of their owners; GHOULBOX is not affiliated with or endorsed by them or by Hügelson Instruments.

# Welcome to the dungeon

GHOULBOX turns the M-VAVE FM-1 into a dungeon-synth instrument: hurdy-gurdies and drones, crypt pads and monk choirs, organs, harps and recorders, all through a cathedral reverb and a worn cassette.

It is a fork of **Felucca**, the open-source FM-1 firmware by Hügelson Instruments. Everything Felucca does still works here: four tracks, a 64-step sequencer, songs, the arpeggiator, chord keys, effects, the web editor and backups. GHOULBOX adds the sounds, the reverb and tape, a hurdy-gurdy engine and an old-RPG look, and leaves out the sounds that don't fit the genre.

## In this guide

- |                              |                           |
|------------------------------|---------------------------|
| 1. Install, back up, go back | 8. Sequencer and patterns |
| 2. Five-minute start         | 9. Saving your work       |
| 3. The panel                 | 10. The menu              |
| 4. The screen                | 11. Dungeon recipes       |
| 5. Pages                     | 12. Troubleshooting       |
| 6. The sounds                | 13. Credits and licence   |
| 7. Effects: hall and tape    |                           |

GHOULBOX is a hobby project, tested on one FM-1. Install it at your own risk. The installer backs up your FM-1 first, and you can always go back to Felucca or M-VAVE's own firmware.

# 1 · Install, back up, go back

## What you need

- A computer with **Chrome or Edge** (other browsers can't talk MIDI to the FM-1)
- A USB **data** cable (some cables only charge)
- No other music software open (it can hold on to the FM-1's MIDI port)

## Install

1. Open [jasonpersinger.github.io/ghoulbox-fm1-dungeon-synth](https://jasonpersinger.github.io/ghoulbox-fm1-dungeon-synth) (it opens the installer at its https address).
2. Connect the FM-1 and press **Install**. Allow MIDI access when the browser asks.
3. The installer saves a **backup** of your music, sounds and settings first, then switches the FM-1 to update mode and writes GHOULBOX.
4. When it says done, the FM-1 restarts with the GHOULBOX skull. Hold **HOME** › SYSTEM › ABOUT shows the version.

If the installer stops after it says "update mode", unplug the USB cable, plug it back in, and press Install again. The FM-1 stays in update mode until the install is finished, so nothing is lost.

## Going back

To return to Felucca, install it from [hugelton.github.io/Felucca](https://hugelton.github.io/Felucca) and restore your backup in its editor. Its **Return to official V15** button puts back M-VAVE's own firmware.

## Your old music

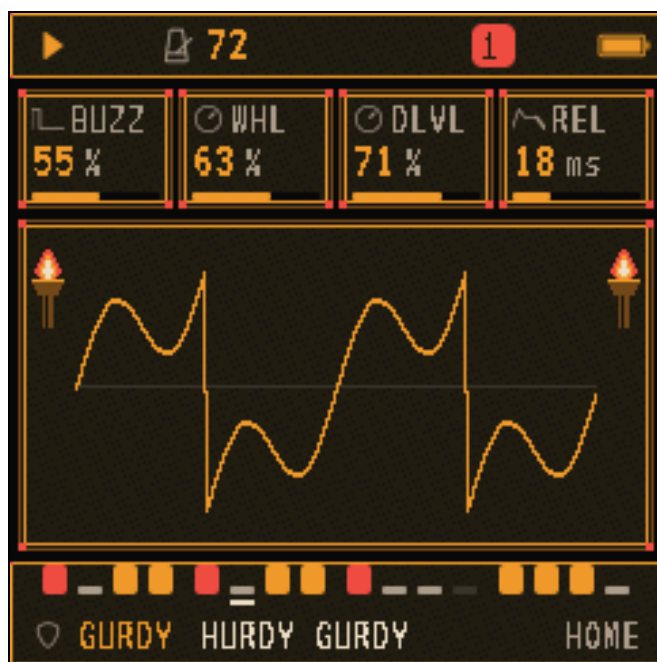
Projects and user presets from Felucca load in GHOULBOX. Sounds GHOULBOX no longer lists (for example the techno presets or the drum kit) still load and play inside an old project; they are only missing from the menus.

## 2 · Five-minute start

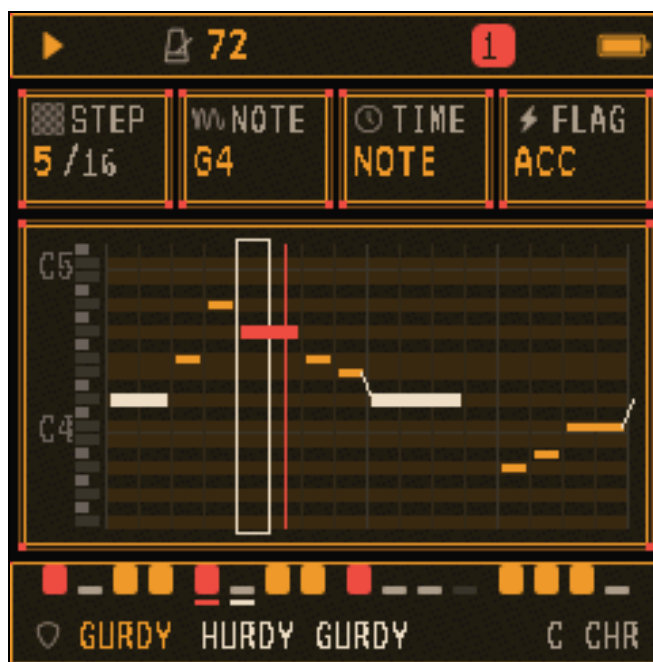
GHOULBOX powers on ready to play a small dungeon piece.

1. Press **PLAY**. Four tracks play a scene in D minor at 72 BPM: a hurdy-gurdy melody (track 1), a crypt pad (2), chanting monks (3) and a drone (4), through the hall reverb. Press PLAY again to stop.
2. Turn **MASTER** for volume. Headphones in the jack silence the speaker.
3. Turn **ALGORITHM** to choose the track (T1–T4). Everything you do applies to that track.
4. Turn **PRESETS** to change the track's sound while it plays. Your pattern stays.
5. Play the keys to play the selected track. **OCT-** / **OCT+** shift the octave.
6. Turn **SELECT** to change the tempo.
7. Turn **KNOB 1-4** to shape the sound: on HOME they are the engine's four main controls.

Made a mess? Hold **SAVE** to undo the last sound or pattern load.



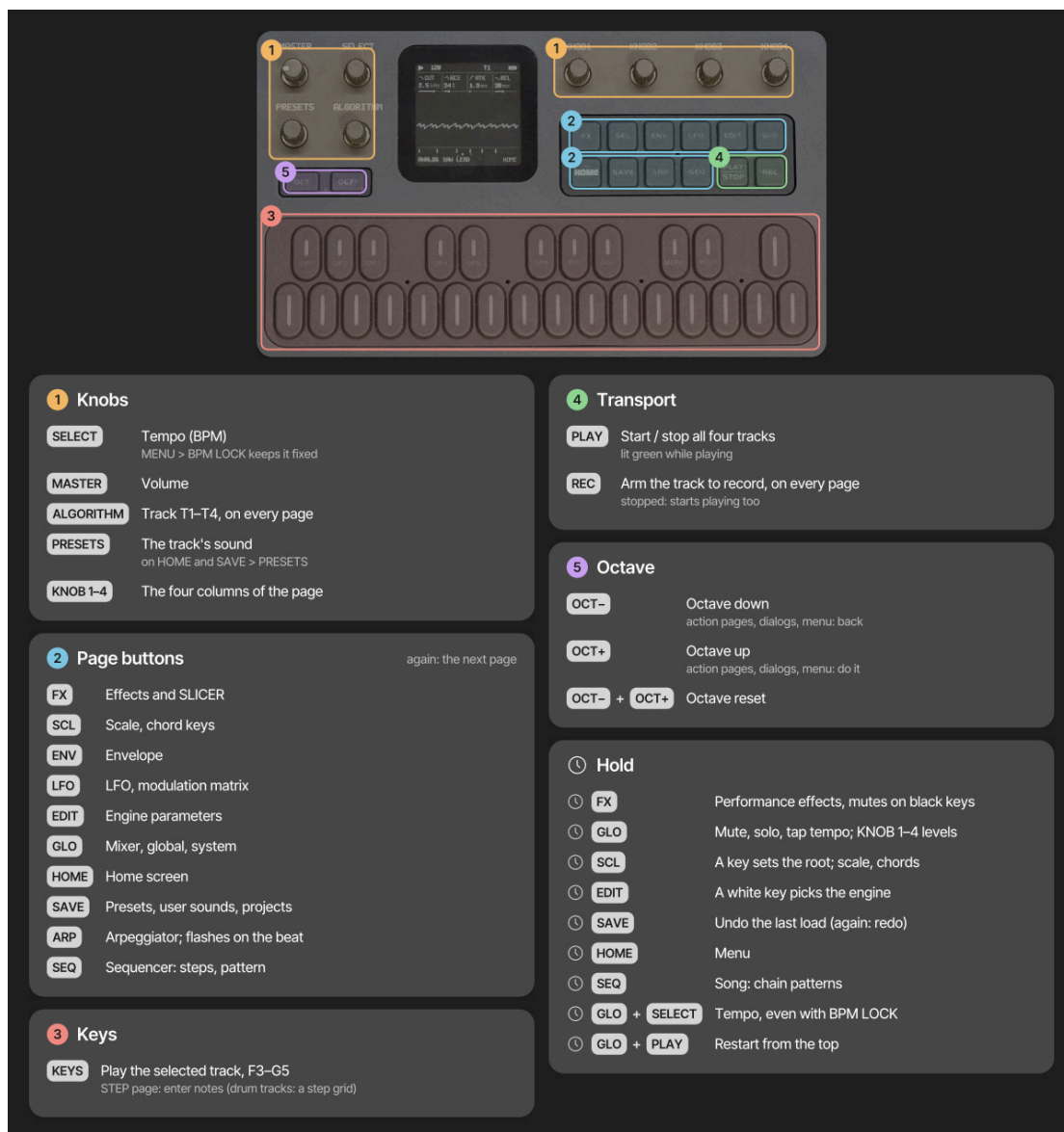
HOME: four knob cards, the scope between the torches, the track's steps and sound at the bottom



SEQ > STEP: the piano roll of the selected track

## 3 . The panel

The FM-1's controls, as Felucca (and GHOULBOX) use them. Tap a page button for its page, tap again for the next page in its group, and **hold** some of them for quick layers.



Panel diagram from Felucca's documentation (Hügelton Instruments, GPL-3.0). The screen in it shows Felucca's look; GHOULBOX's is on the next page.

## Quick layers

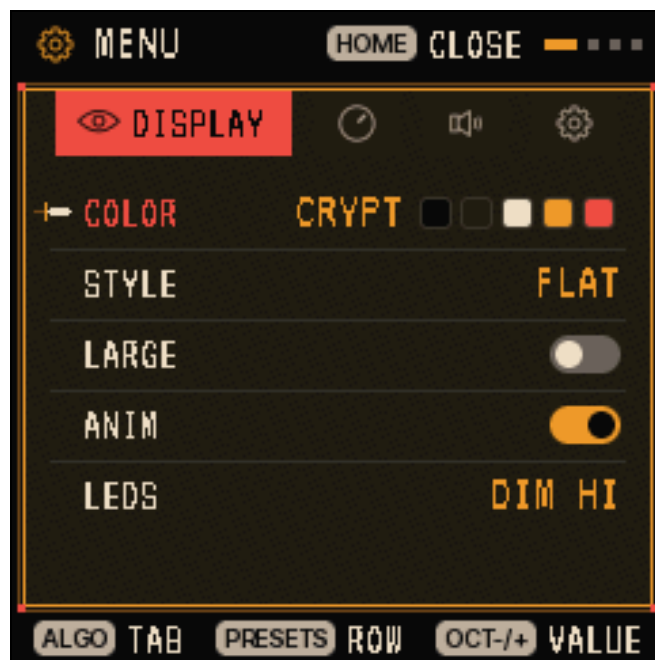
Hold **FX**, **GLO**, **SCL** or **EDIT** to put shortcuts on the keys and knobs; the screen shows the map. Double-tap one to keep its layer open; tap again to close it. In the **EDIT** layer, each white key picks an engine. **OCT+** confirms and **OCT-** goes back, on every action page and dialog.

## 4. The screen

GHOULBOX draws Felucca's pages as an old role-playing game would: stone windows, a pixel font, flickering wall torches beside the scope, and a **sword** that points at what is selected.



PRESETS: the sword marks the current sound



The MENU (hold HOME) in four tabs

- **Colours:** a new FM-1 starts in **CRYPT** (stone, bone, torch amber, blood red). If you came from Felucca, your saved palette stays: hold **HOME** and pick COLOR › CRYPT.
- **The header** shows play state, tempo, the track number and battery.
- **Knob cards** show the four values the knobs change; the one you just turned is drawn in the accent colour.
- **Messages** (for example SOUND INIT or NO SAMPLE) appear in the header for a moment.

# 5 · Pages

Each page button opens a group of pages; tap it again for the next. On every page, **Knob 1–4** change the four values shown.

Button Pages (tap again for the next)

|             |                                                                                                                            |
|-------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>HOME</b> | HOME: the engine's four main controls and the scope                                                                        |
| <b>ENV</b>  | ENV (attack, decay, sustain, release) · ENV DEST                                                                           |
| <b>LFO</b>  | LFO · LFO DEST · MOD (the modulation matrix)                                                                               |
| <b>FX</b>   | FX (the track's sends: distortion, chorus, delay, reverb) · SLICER · DLY · REVERB · CHORUS · TAPE                          |
| <b>SCL</b>  | SCL (root, scale, white-key mode, transpose) · CHORD                                                                       |
| <b>EDIT</b> | The engine's own parameters (two pages of four) · VOICE (poly, mono, legato, unison, glide) · SLICES (on the SLICE engine) |
| <b>GLO</b>  | MIXER · GLOBAL (tempo, swing, clock, tune) · SYSTEM (MIDI and sync)                                                        |
| <b>SAVE</b> | PRESETS · USER (your 32 sounds) · PROJECT (4 projects) · TOOLS                                                             |
| <b>ARP</b>  | ARP (arpeggiator)                                                                                                          |
| <b>SEQ</b>  | STEP (piano roll) · PATTERN · PHRASES (load a factory pattern) · CHANCE · AUTOMATION; hold SEQ for SONG                    |

The PRESETS knob changes the sound on every page that is not a list, so you can audition sounds from anywhere. On the SEQ pages it moves the step cursor instead.

# 6 · The sounds

57 sounds across 12 engines. Hear them all at

[jasonpersinger.github.io/ghoulbox-fm1-dungeon-synth/sounds](https://jasonpersinger.github.io/ghoulbox-fm1-dungeon-synth/sounds). GB marks GHOULBOX's own; the rest are Felucca's that fit the genre.

| Engine                                | Sounds (in PRESETS order)                                                                                              |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>ANALOG</b><br>virtual analog       | SOFT PAD · PWM STR · SUB BASS · WIND · STRINGS · CRYPT PAD GB · FROST STR GB · WAR HORN GB · DIRGE BASS GB             |
| <b>FM6</b><br>6-operator FM           | BELL · PAD · FRENCH HORN GB · VIRGINAL GB · GLASS CHOIR GB · DARK STRINGS GB · LOFI FLUTE GB · PIPE ORGAN GB · TUBA GB |
| <b>PHASE</b><br>phase distortion      | STRING · GRIM BRASS GB                                                                                                 |
| <b>LOFI</b><br>chiptune               | TOWER LEAD GB · CASIO CHOIR GB · RECORDER GB                                                                           |
| <b>SAMPLE</b><br>real recordings      | BOWED PSALT GB · REN ORGAN GB · TENOR RECORD GB · FOLK HARP GB                                                         |
| <b>VOICE</b><br>sung vowels           | CHOIR AAH · MONKS GB · CRYPT CHOIR GB · WRAITH GB · LOW DRONE GB                                                       |
| <b>TRIO</b><br>three oscillators      | CHIP CHOIR · FANTASY PAD GB · MOURN HORN GB · STRING MACH GB · SHAWM GB                                                |
| <b>WHEEL</b><br>tonewheel organ       | FULL ORGAN · SOFT FLUTE · CRYPT ORGAN GB · CATHEDRAL GB · HARMONIUM GB · TOWER FLUTE GB                                |
| <b>GRAIN</b><br>granular              | CLOUD PAD · FROZEN                                                                                                     |
| <b>PHYS</b><br>physical models        | LUTE GB · HARPSICHORD GB · DUNGEON HARP GB · GURDY DRONE GB                                                            |
| <b>GURDY</b><br>hurdy-gurdy           | HURDY GURDY GB · DRONE WHEEL GB · DANCE GURDY GB · TROMPETTE GB · VIELLE GB · BAGPIPE GB                               |
| <b>NOISE</b><br>noise and<br>ambience | CAVE WIND GB · TORCH GB                                                                                                |

## Real recordings

The SAMPLE engine plays four real instruments, recorded by Versilian Studios and released into the public domain (CC0): a **bowed psaltery** (a medieval zither played with a bow), a **Renaissance chamber organ**, a **baroque tenor recorder** and a **folk harp**. Held notes on the first three sustain for as long as you hold the key. SAMPLE's SET knob lists them first, then your own samples (USR1–USR3, uploaded from the web editor).

## The hurdy-gurdy (GURDY)

A hurdy-gurdy plays one melody note at a time over drones that never stop. Its eight controls (**EDIT**):

| Control     | What it does                                                                     |
|-------------|----------------------------------------------------------------------------------|
| <b>DRN</b>  | The drone's note (C–B): set it to your song's key                                |
| <b>DLVL</b> | Drone level                                                                      |
| <b>5TH</b>  | A second drone a fifth above, and its level                                      |
| <b>WHL</b>  | Wheel pressure: darker when low, brighter and rougher when high                  |
| <b>BUZZ</b> | The trompette, a buzzing bridge: how hard it rattles                             |
| <b>COUP</b> | When it buzzes: always (HOLD) or rhythmic strokes in time (1/4, 1/8, 1/8T, 1/16) |
| <b>BODY</b> | The wooden body's resonance                                                      |
| <b>WOBL</b> | The wheel's unevenness: a slow wander in pitch and loudness                      |

## 7 · Effects: hall and tape

Each track sends to shared effects; their amounts are on the **FX** page (KNOB 1–4: distortion, chorus, delay, reverb). Tap **FX** again to reach the effects themselves.

### REVERB

- **TYPE** (KNOB 1): ROOM, SPRING or **HALL**. GHOULBOX starts in HALL.
- **SIZE**: from a 1.5-second stone chamber to a 12-second cathedral.
- **DAMP**: darkens the tail.

### TAPE

- **TAPE**: the whole mix through a worn cassette: wow and flutter (wavering pitch), saturation, treble loss and hiss. GHOULBOX starts at 40.
- **CRSH**: a cheap sampler: fewer bits and a lower sample rate. Off by default; a little adds grit, a lot turns to crunch.

### DLY and CHORUS

A delay (time, feedback, colour, mix) and a chorus (rate, depth). With HALL selected, the delay is limited to 0.74 s, because the hall borrows part of its memory.

### The FX layer

Hold **FX** to play performance effects from the keys (repeat, reverse, filter sweeps, tape stop, freeze, a harmonizer) and mute tracks with the black keys. With MENU › FX LATCH on, they toggle instead of needing to be held.

## 8 · Sequencer and patterns

Each track has its own sequence of up to 64 steps, with chords, ties, accents, slides, per-step chance and ratchets.

### Load a factory pattern

**SEQ** › PHRASES lists ready-made patterns to load into the selected track. GHOULBOX adds eight in D minor, over the power-on scene's chords: **DIRGE** (slow pad chords), **BALLAD** (a melody), **CHANT** (low voices) and **PEDAL** (one held low D), which the power-on scene uses, and **MARCH** (a stepping bass), **LAMENT** (a falling melody), **ARPEGGIO** (broken chords for harp or lute) and **ANTIPHON** (a call answered an octave lower). Loading a sound never changes your pattern, and loading a pattern never changes your sound.

### Write your own

- **Step entry:** on **SEQ** › STEP, move the cursor with PRESETS and play a key to write it into the step.
- **Live recording:** press **REC** to arm the track (from any page), then play while the sequence runs; it loops and overdubs.
- **Length and speed:** SEQ › PATTERN sets the length (LEN) and step size (DIV, from 4 bars down to 1/32). The power-on scene uses 1/8 steps for its slow pace.
- **Clear:** hold **REC** on a SEQ page; **OCT+** clears, **OCT-** cancels.

### Scales and chords

On **SCL**, set QNT to WHITE to play the chosen scale on the white keys only. For dungeon synth, natural minor (MIN), Dorian (DOR), Phrygian (PHRY) and harmonic minor (HARM) are all at home. Tap SCL again for CHORD: one finger then plays a chord of the scale.

### Songs

Hold **SEQ** for SONG, to chain patterns A–D into a longer piece.

## 9 . Saving your work

- **Save a sound** (32 user slots): stop, tap **SAVE**, pick a slot with KNOB 1, press **OCT+**, name it with the keys, and press **OCT+** again.
- **Projects** (4): **SAVE** › PROJECT keeps the whole session: all four tracks, their sounds and sequences, songs and settings.
- **Undo**: hold **SAVE** to go back before the last sound or pattern load (hold again to redo).
- **Backups**: the web editor (on the installer site) makes a full backup to your computer and restores it.

The FM-1 always starts with the dungeon scene. Your saved projects stay in their slots; load one from SAVE › PROJECT.

## 10 · The menu

Hold **HOME** for the menu. **ALGORITHM** moves between its four tabs, **PRESETS** through the rows, and any knob or **OCT-** / **OCT+** changes a value. Press **HOME** to close it.

| Tab     | Settings                                                                                                                                                                                       |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DISPLAY | <b>COLOR</b> (palette; GHOUlBOX's is CRYPT) · <b>STYLE</b> (FLAT or LINE) · <b>LARGE</b> (bigger knob cards) · <b>ANIM</b> (rolling digits on or off) · <b>LEDS</b> (OFF, DIM LO, DIM HI, INV) |
| CONTROL | <b>HOLD</b> (how long a quick layer takes to open) · <b>KNOB ACCEL</b> · <b>FX LATCH</b> · <b>BPM LOCK</b> (SELECT stops changing the tempo)                                                   |
| AUDIO   | <b>SPEAKER EQ</b> (FLAT, LOWCUT, BASS+; keep FLAT when recording) · <b>USB LEVEL</b> (MASTER or FIXED)                                                                                         |
| SYSTEM  | <b>USB SERIAL</b> (turn OFF on macOS 13–15 so the USB audio input appears) · <b>HARDWARE CALIBRATION</b> · <b>ABOUT</b> (the version)                                                          |

### Recording to a computer

The FM-1 appears on a computer as a USB audio input named "Felucca", carrying the master output at 44.1 kHz with no driver. Set **SPEAKER EQ** to **FLAT**, and **USB LEVEL** to **FIXED** if you want a level that doesn't follow the **MASTER** knob.

# II · Dungeon recipes

Starting points. Load the sound with PRESETS, then adjust.

## Old Tower-style pad

CRYPT PAD or DARK STRINGS on a track, the DIRGE pattern from SEQ › PHRASES, REVERB SIZE high, TAPE around 60. Slow everything down: tempo 60–70, DIV 1/4 or 1/2.

## Lone hurdy-gurdy

HURDY GURDY with DRN set to your key. Use COUP 1/8 for rhythmic buzz strokes on a dance tune, or BUZZ at zero for a sad, plain drone.

## Monastery

MONKS on one track and CRYPT CHOIR an octave above on another, the CHANT pattern, REVERB HALL with SIZE at maximum. Add PIPE ORGAN or REN ORGAN holding the root.

## Tavern ballad

FOLK HARP or LUTE on an arpeggio (ARP), TENOR RECORD or LOFI FLUTE for the tune, BAGPIPE for a reedy lead over its drones.

## Lo-fi cassette demo

Any of the above with TAPE 80–100 and CRSH 30–50: the sound of a 1990s dungeon-synth tape.

## 12 · Troubleshooting

| Problem                                         | What to try                                                                                                                                                                                                    |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No sound                                        | Turn up MASTER; check the track isn't muted (GLO › MIXER); headphones in the jack silence the speaker.                                                                                                         |
| The installer can't find the FM-1               | Use a USB data cable, close other music software, use Chrome or Edge.                                                                                                                                          |
| The installer stops after "update mode"         | Unplug and replug the USB cable and press Install again. The FM-1 waits in update mode until finished.                                                                                                         |
| "NO SAMPLE" and a plain tone                    | The SAMPLE or GRAIN source is an empty user slot (USR1–3). Pick a built-in set or upload a sample in the web editor.                                                                                           |
| No USB audio on a Mac                           | MENU › SYSTEM › USB SERIAL OFF (macOS 13–15).                                                                                                                                                                  |
| The screen is black after an interrupted update | If the computer sees a device named WL80UBOOT, the FM-1 is in its boot mode and can be recovered: try another cable and run the installer again. Felucca's README describes further recovery.                  |
| Something else                                  | Report it at <a href="https://github.com/jasonpersinger/ghoulbox-fm1-dungeon-synth/issues">github.com/jasonpersinger/ghoulbox-fm1-dungeon-synth/issues</a> with your OS, browser and the installer's log text. |

# 13 · Credits and licence

GHOULBOX © 2026 Jason Persinger. Free software under the **GPL-3.0-only** licence; the source is at [github.com/jasonpersinger/ghoulbox-fm1-dungeon-synth](https://github.com/jasonpersinger/ghoulbox-fm1-dungeon-synth).

Built entirely on **Felucca** by Leo Kuroshita / Hügelton Instruments (GPL-3.0-only). Please support the original: GitHub Sponsors (hugelton) or itch.io (hugelton.itch.io/felucca).

- Samples: Versilian Studios, VSCO-2 Community Edition and VCSL (CC0 1.0)
- Fonts: VT323 by Peter Hull and Pirata One by Rodrigo Fuenzalida and Nicolas Massi (SIL OFL 1.1)
- Felucca's engines draw on Dexed (FM6), DaisySP and Mutable Instruments' code (PHYS), klattsch (VOICE) and CrispyZebra (PHASE); see Felucca's README for the full credits

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